Draft version November 29, 2025

Typeset using LaTeX **twocolumn** style in AASTeX631

Numerical Simulation of Two-Body Orbits Using Runge-Kutta Methods

Lily Nguyen[1] and Andre Sae[1]

[1] *Department of Physics, The University of Texas at Austin*
Austin, TX 78712, USA

ABSTRACT

We numerically solve the two-body gravitational problem using the fourth-order Runge-Kutta method to study how mass ratio and orbital eccentricity affect orbital motion. After reducing the system to an effective one-body problem, we simulated 16 configurations consisting of four mass ratios ($m_1 : m_2 = 1 : 1, 1 : 2, 1 : 4, 1 : 16$) and four eccentricities ($\varepsilon = 0, 0.25, 0.5, 0.75$). We set initial conditions to ensure closed orbits and constructed the full trajectories of both bodies in the center-of-mass frame. We found that increasing the eccentricity led to more elongated orbits and higher velocities near periapsis, which is consistent with the conservation of angular momentum. As the mass ratio became more extreme, the heaver body remained close to the center of mass while the ligher one followed a significantly larger orbit, which aligns with theoretical expectations and highlights the effectiveness of numerical methods in modeling classical gravitational systems. This framework can be extended to include perturbations, spin-orbit coupling, or additional bodies, which are relevant to modern applications in astrophysics.

Keywords: two-body problem — orbital dynamics — Runge-Kutta method — mass ratio — eccentricity — numerical simulation

1. INTRODUCTION

The two-body problem from classical mechanics occurs when two masses interact through a central force, most commonly gravity. It appears in physical systems, such as the Earth orbiting the Sun, binary star pairs, and the motion of artificial satellites. Despite its simplicity, this problem captures essential features of orbital motion and serves as a foundation for more complex models. Because it can be solved exactly within the framework of Newtonian mechanics, it holds both theoretical and practical importance in physics and astronomy Murray & Dermott (2012).

Given two masses, m_1 and m_2, interacting via Newton's laws of graviation, the force on each body is governed by:

$$\vec{F} = -G\frac{m_1 m_2}{r^2}\hat{r}, \tag{1}$$

where G is the gravitational constant, r is the distance between the two masses, and $\hat{r}$ is the unit vector pointing from one mass to the other Taylor (2005).

To simplify our analysis, we can reduce the two-body system to an effective one-body problem by introducing the relative position vector $\vec{r} = \vec{r}_1 - \vec{r}_2$ and the reduced mass $\mu = \frac{m_1 m_2}{m_1+m_2}$. The resulting equation of motion becomes:

$$\mu\ddot{\vec{r}} = -G\frac{m_1 m_2}{r^2}\hat{r}, \tag{2}$$

which describes the motion of a single particle of mass μ undergoing a central force. The general solution to this equation yields conic-section trajectories (elliptical, parabolic, or hyperbolic) depending on the total energy and angular momentum of the system Taylor (2005).

The two-body problem can be solved analytically, which highlights its importance in understanding more complex dynamical systems such as celestial mechanics, orbital determination, and pertubation theory Meyer et al. (2023); Schneider et al. (2011). In computational contexts such as this project, it also provides a useful model for implementing and validating numerical methods like the Runge-Kutta algorithm Sullivan (2022).

In this project, we observe how variations in mass ratios and orbital eccentricity affect the resulting orbits in the two-body problem. We numerically solve the equations of motion using the fourth-order Runge-Kutta method and visualize the resulting trajectories under these various configurations.

2. DATA AND OBSERVATIONS

The two-body problem involves two point masses, m_1 and m_2, interacting via a central force. To simplify, we can reduce it to an equivalent one-body problem by switching from absolute position vectors $\vec{r}_1$ and $\vec{r}_2$ to the relative position vector $\vec{r} = \vec{r}_1 - \vec{r}_2$.

By applying Newton's second law to each body, the graviational force acting on m_1 due to m_2 becomes:

$$\vec{F}_{12} = -G\frac{m_1 m_2}{|\vec{r}_1 - \vec{r}_2|^2}\hat{r}, \tag{3}$$

and the force acting on m_2 is simply $\vec{F}_{21} = -\vec{F}_{12}$. Subtracting the two equations of motion eliminates forces gives us an effective second-order differential equation for the relative coordinate:

$$\mu\ddot{\vec{r}} = -G\frac{m_1 m_2}{r^2}\hat{r}, \tag{4}$$

where $\mu = \frac{m_1 m_2}{m_1+m_2}$ is the reduced mass of the system and $r = |\vec{r}|$.

This equation describes the motion of a single particle of mass μ in a central graviational potential, and the resulting orbits are conic sections (ellipses, parabolas, or hyperolas) depending on the system's energy and angular momentum Taylor (2005); Murray & Dermott (2012).

Once we solve for the relative motion, we can recover the original trajectories using the center-of-mass frame:

$$\vec{r}_1 = \frac{m_2}{m_1+m_2}\vec{r}, \quad \vec{r}_2 = \frac{m_1}{m_1+m_2}\vec{r}. \tag{5}$$

To study the dynamics of the reduced one-body system, we placed the particle of reduced mass at $(x_0, y_0) = (-1, 0)$ with an initial velocity $(v_{x0}, v_{y0}) = (0, \sqrt{1+\varepsilon})$ and constrained $\varepsilon < 1$ to ensure closed-bounded orbits. We evolved the system over one orbital period:

$$T = \frac{2\pi}{(1-\varepsilon)^{3/2}} \tag{6}$$

using four values for eccentricity $\varepsilon = (0, 0.25, 0.5, 0.75)$ and four mass ratios $m_1 : m_2 = (1:1, 1:2, 1:4, 1:16)$, totaling 16 configurations.

We solved the resulting system of first-order ODEs using the fourth-order Runge-Kutta method implemented via `solve_ivp` from Python's `SciPy` library with 500 time steps per orbit Sullivan (2022).

3. RESULTS

After obtaining the relative trajectories, we reconstructed the positions of the two bodies using Equation 5. The resulting orbital trajectories are shown in Figure 1.

4. SUMMARY AND CONCLUSION

In this project, we numerically analyzed the two-body problem using the fourth-order Runge-Kutta method to solve the equations for 16 distinct configurations. We observed how changing the mass ratio and orbital eccentricity parameters influenced the shape and dynamics of the resulting orbits. As expected, increasing the eccentricity led to more elongated trajectories and sharper velocity changes near periapsis, which aligns with the conservation of angular momentum Murray & Dermott (2012).

We also found that greater mass asymmetry causes the more massive body to remain closer to the system's center of mass while the ligher body follows a larger orbit. In the limiting case of a very large mass ratio, the heavier object remains nearly fixed. Physically, the behavior of a small body orbiting a much larger one is in line with a planet orbiting a star. Altogether, our results agree with theoretical predictions and highlight the effectiveness of numerical methods in modeling gravitational systems beyond simple analytic cases.

Real-world uses of the two-body problem highlight its ongoing relevance in modern astrophysics. For instance, NASA's DART mission used perturbed two-body dynamics to model post-impact changes in the Didymos-Dimorphos binary system to help estimate orbital shifts and assess the effectiveness of asteroid deflection strategies Meyer et al. (2023). Similarly, accurate two-body and N-body models are essential in characterizing and cataloging exoplanetary systems, such as those in the exoplanet.eu database Schneider et al. (2011).

Overall, these applications demonstrate how even a well-understood classical system like the two-body problem continues to influence active areas of research such as as-

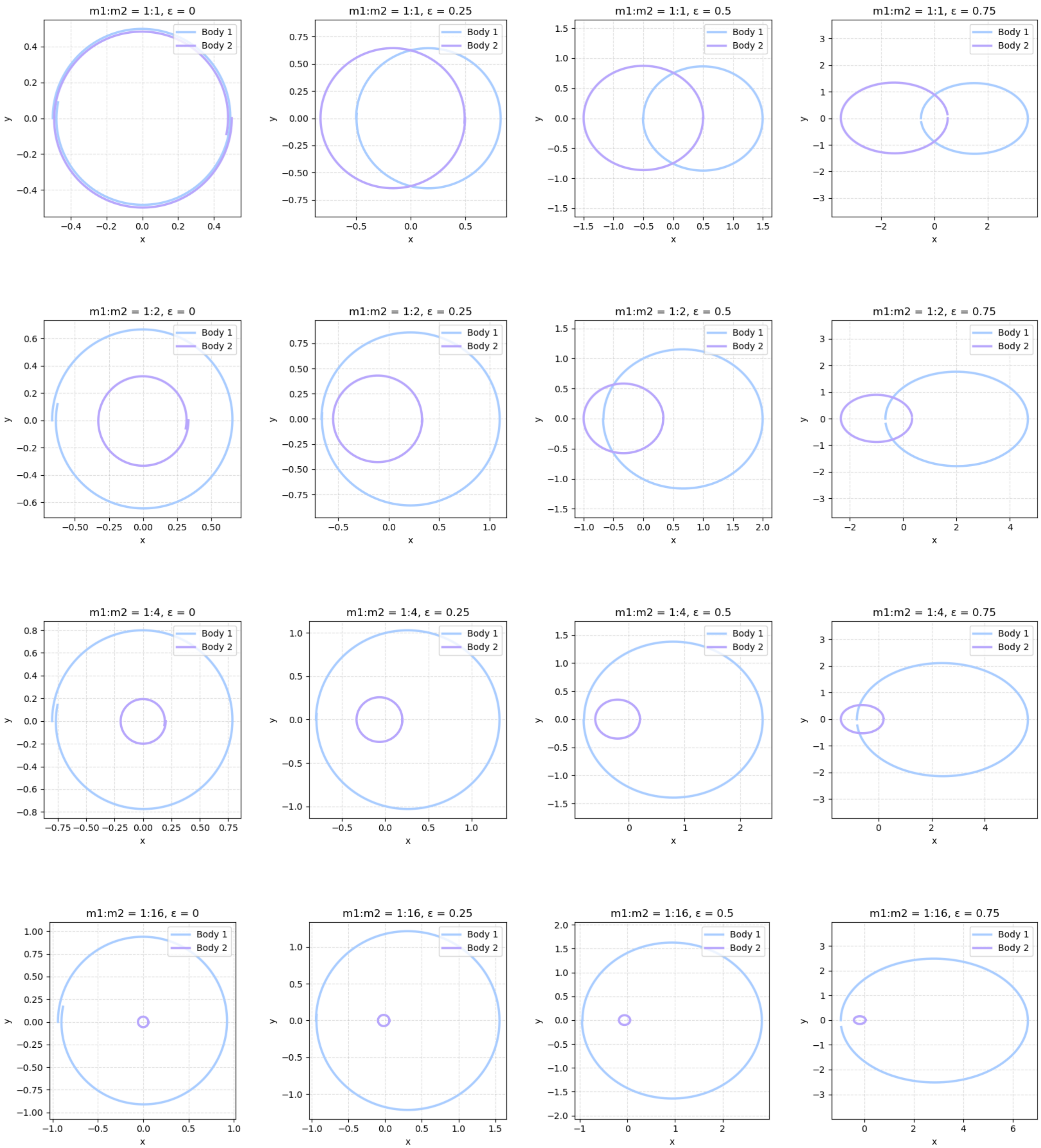

Figure 1. Two-body orbit plots for various mass ratios ($m_1 : m_2$, top to bottom) and eccentricities (ε, left to right).

teroid deflection strategies and exoplanet discovery and classification Meyer et al. (2023). Further extensions could include modeling perturbative forces (e.g. non-gravitational effects or third-body influences), validating simulations against real astronomical data, or exploring transitions to chaotic behavior. These additions would bridge the gap between theory and the complexities of real-world orbital systems.

ACKNOWLEDGEMENTS

We thank Professor Mitra for his guidance and instruction throughout the course, as well as the teaching assistants for their helpful feedback. We also acknowledge the Department of Physics for providing an excellent academic foundation to support this work. This work was completed as part of the requirements for C S 323E: Elements of Scientific Computing at the University of Texas at Austin.

REFERENCES

Meyer, A. J., Agrusa, H. F., Richardson, D. C., et al. 2023, The Planetary Science Journal, 4, 141, doi: 10.3847/PSJ/acebc7

Murray, C. D., & Dermott, S. F. 2012, The Two-Body Problem (Cambridge: Cambridge University Press), 22–62, doi: 10.1017/CBO9781139174817

Schneider, J., Dedieu, C., Le Sidaner, P., Savalle, R., & Zolotukhin, I. 2011, Astronomy and astrophysics (Berlin), 532, A79

Sullivan, E. 2022, Numerical Methods: An Inquiry-Based Approach with Python (Available at numericalmethodssullivan.github.io). https://numericalmethodssullivan.github.io/

Taylor, J. R. 2005, Classical mechanics / John R. Taylor. (Sausalito, Calif: University Science Books)